Departamento de
Economía
Universidad de Concepción

# The Impact of Environmental Variability on Fishers' Harvest Decisions in Chile

## A multi-species structural approach

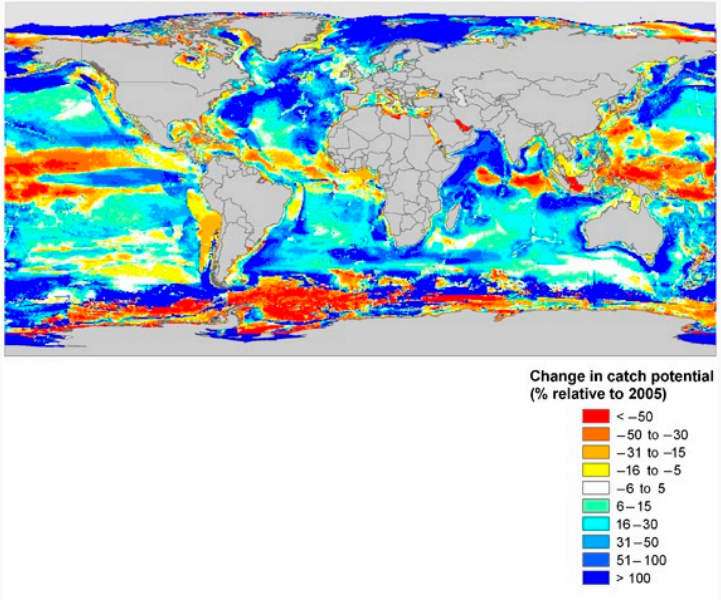
Felipe J. Quezada-Escalona

ICES/PICES Small Pelagic Symposium — La Paz, México, May 8, 2026

# Motivation



- Small pelagic stocks fluctuate sharply with climate variability.
- **Standard projection tools assume constant fishing rates or full quota uptake** — the human decision is assumed away.
- But fishers **choose**: which species to target, how many trips, how much to land.




**Source:** Cheung et al. (2010)

> ***Project goal:*** *propagate climate-driven biomass shifts into fisher-side responses — participation today (Paper 1),* ***optimal harvest decisions*** *in the full structural model (Paper 2).*

# The project — two papers, one structural pipeline



**Paper 1** (today's results) *Reduced-form effort + identified stock dynamics*

- **Negative binomial** trip equation, estimated separately by fleet.
- Bayesian state-space stock dynamics with structural climate shifters (in appendix).
- Comparative statics under future climate scenarios.

**Paper 2** (today's roadmap) *Welfare counterfactuals*

- Translog **cost** function (trip-level, by fleet).
- **Inverse demand** (IAIDS, 3SLS).
- **Dynamic vessel optimization** $\Rightarrow$ optimal $T_g, h_{g\tau}$ under different climate scenarios.
- Welfare, prices, **species substitution**.

> *The two papers share the **same identified stock dynamics**. Paper 1 stops at participation; Paper 2 closes the loop through prices, cost and profits to estimate optimal harvest and quota allocation.*

# Setting and observed heterogeneity

# Chile's Small Pelagic Fishery (Centro-Sur)



## Stocks & sectors

- **Anchoveta** (*Engraulis ringens*), **sardine** (*Strangomera bentincki*), **jack mackerel** (*Trachurus murphyi*).
- ~94% of national landings; mostly purse seiners.
- Two sectors: **artisanal** (ART), **industrial** (IND).

## Quota regime

- TAC split between sectors with **limited cross-sector transferability**.
- IND quotas partially tradable within sector; ART under spatial collective-property rights (i.e. **RAE**).

## Fleet portfolios are very different

- **ART**: ~ {anchoveta, sardine}. Concentrated.
- **IND**: ~ 95% jack mackerel + 5% sardine.

## Mobility

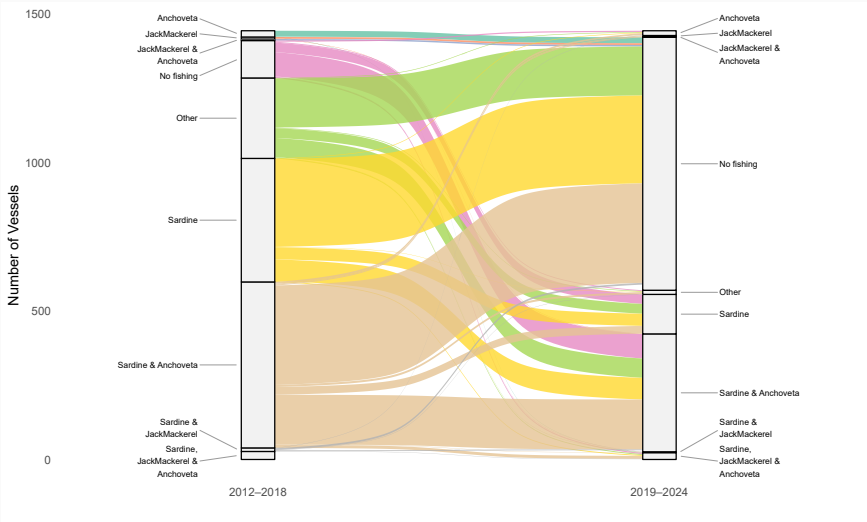
- IND: offshore purse-seiners.
- ART: port-anchored, RAE binds spatially.

$\Rightarrow$ Same shock, **different choices**.

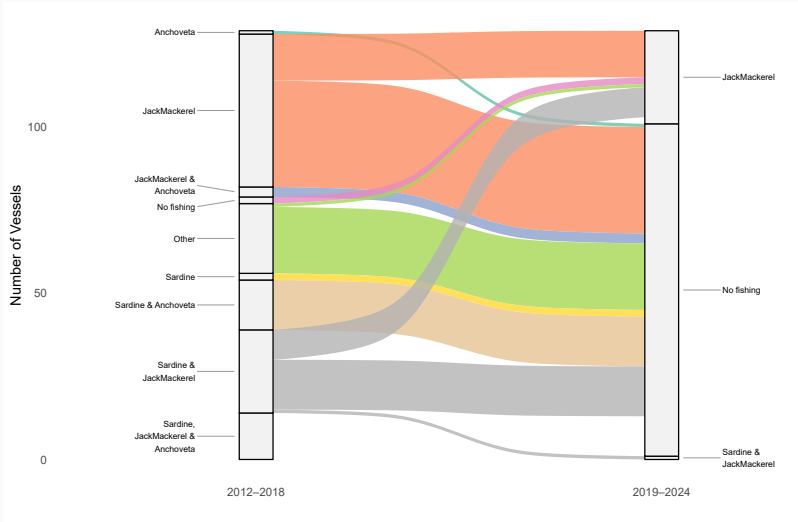
# Observed strategy switching — ART vs IND

Departamento de Economía
Universidad de Concepción

## Artisanal




## Industrial




Trip-level target-species transitions, IFOP logbooks 2013–2024.

> *ART: anchoveta↔sardine dominates; jack mackerel essentially absent. IND: rotation across all three. Same fishery, very different revealed choice sets.*

# Paper 1 — The NB trip equation

# A negative binomial model of annual trips



## Specification, by fleet $g$ (ART, IND)

$$T_{vy}^{g} \sim \mathrm{NB}\big(\exp(U_{vy}'\beta_g + \alpha_y^g),\ \theta_g\big)$$

with $U_{vy} = \big[\, p_{sy},\ H_{vy}^{alloc},\ Z_v,\ O_{vy} \,\big]$:

- $p_{sy}$: ex-vessel prices, by species.
- $H_{vy}^{alloc}$: vessel-level quota allocation.
- $Z_v$: hold capacity, vessel type.
- $O_{vy}$: bad-weather days, closures.
- $\alpha_y^g$: **year FE** (absorbs 2019–22 shocks).
- Vessel-clustered robust SEs.

## Why NB by fleet?

- Overdispersion $\sigma^2/\mu$: **22.4** (ART), **4.9** (IND). Poisson rejected at $p < 0.001$.
- Two fleets, two technologies, two choice sets — no pooling.

## Two channels to projections

- **Indirect**: structural shifters $\rho_i^{SST}, \rho_i^{CHL} \to B_{i,t} \to H_{vy}^{alloc}$.
- **Direct**: vessel-specific $\Delta$bad-weather days from CMIP6 winds.

# What drives fisher participation?

Departamento de Economía
Universidad de Concepción

| Variable | Industrial | Artisanal | Reads as |
|---|---|---|---|
| $H_{vy}^{alloc}$ | **(+)** *** | **(+)** *** | Quota = binding budget constraint, both fleets. |
| Price — jack mackerel | **(+)** *** | 0 | IND oriented to jack mackerel; ART has no exposure. |
| Price — sardine | 0 | **(+)** *** | Sardine is ART's primary target. |
| Price — anchoveta | 0 | **(–)** *** | Simultaneity sign. |
| Hold capacity (log) | 0 | **(+)** *** | Heterogeneity is *within* the artisanal fleet. |
| Bad-weather days | 0 | **(–)** *** | ART vulnerable; IND insulated by vessel size. |
| Closure days | **(–)** *** | **(+)** ***† | IND: largest single coef. ART: zone FE collinear. |

Negative binomial **with year FE**, vessel-clustered robust SEs. Sign and significance shown. * $p < 0.01$. † zone-FE collinear (not causally interpretable).

> *Two fleets, **completely different participation models**.*

# Comparative statics — the fleet asymmetry

Departamento de Economía
Universidad de Concepción

| Fleet | Counterfactual | $\%\Delta T$ marginal | $\%\Delta T$ conditional | $\Pr(\text{portfolio loss} > 50\%)$ |
|---|---|---|---|---|
| **Artisanal** | mid-century, moderate (SSP2-4.5) | **–8.1%** | –2.1% | **0.95** |
| | end-century, severe (SSP5-8.5) | **–10.2%** | –2.7% | **0.99** |
| **Industrial** | mid-century, moderate (SSP2-4.5) | –0.7% | –0.7% | 0.12 |
| | end-century, severe (SSP5-8.5) | –1.1% | –0.8% | 0.12 |

$\%\Delta T$: change in annual trips vs baseline. **Marginal** averages over all posterior draws. **Conditional** restricts to draws with $f_v^H > 0.5$. Cross-median across a 6-model CMIP6 ensemble. NB with year FE.

- Marginal asymmetry ~**11:1** ART:IND; conditional ~**3.4:1**.
- ART portfolio collapse near-certain by SSP5-8.5 end.

- $\Rightarrow$ Asymmetry runs through **probability of portfolio collapse** $\times$ **direct weather** ($\beta_w^{ART} < 0$, $\beta_w^{IND} = 0$); quota cross-sector limits lock it in.
- IND's protection rests on jack mackerel (long-run shifter **n.i.**).

# Two channels: direct vs indirect



## Decomposition

**Indirect** (biomass → allocation):

$$\Delta SST,\ \Delta \log CHL \xrightarrow{\rho_i^{SST},\ \rho_i^{CHL}} B^{\star}_{i,t} \rightarrow H^{alloc}_{vy}$$

**Direct** (weather → trips):

$$\Delta\text{wind} \xrightarrow{\text{vessel CDF}} \Delta\text{bw-days}_{vy} \xrightarrow{\beta^g_w} T_{vy}$$

> *Indirect (biomass) carries **most** of the asymmetry; direct (weather) reinforces it where vessels are exposed (ART). Both channels point the same way.*

## End-century SSP5-8.5

| Fleet | Indirect | Direct | Total |
|---|---|---|---|
| **Artisanal** | −9.0 pp | −1.2 pp | **−10.2%** |
| **Industrial** | −1.0 pp | −0.1 pp | **−1.1%** |

# Paper 2 — closing the loop

# Three structural extensions ⇒ welfare



## What gets added on top of Paper 1

**(i) Trip-level cost function** — translog $C_{vg}(w, h, x, Z, Env)$, SUR by fleet. Recovers **input demands** (fuel, labor, ice) and elasticities of input substitution under climate shocks.

**(ii) Inverse demand (IAIDS)** — $\ln p_{iy}$ as a function of all species' landings, instrumented by SST, CHL, fuel prices. Resolves the simultaneity observed in paper 1.

**(iii) Vessel dynamic optimization** — $\max_{T_g, h_{g\tau}} \sum_\tau \delta^\tau [P(h)\, h - C_g(h)]$ s.t. quota cap. $(T_g^*, h_{g\tau}^*)$ under each scenario.

## Optimization model allows for

**(i) Prediction under a changing climate of:** Equilibrium prices and species substitution

**(ii) Define optimal quota allocation**

**(iii) Study policy reforms (i.e., contrafactuals) as adaptation:**

- Cross-sector quota transferability.
- RAE redesign (multi-zone permits).
- State-contingent TACs.

# Takeaways



- **Paper 1**: fleet-specific NB model of annual trips reveals **two completely different participation technologies** in the same fishery.
- **Paper 1**: under counterfactual shocks, the marginal ART:IND participation asymmetry is **~11:1** (conditional **~3.4:1**), running through (i) the **probability of portfolio collapse** in the indirect biomass channel and (ii) **direct weather** exposure ($\beta_w^{ART} < 0$, $\beta_w^{IND} \approx 0$); quota cross-sector limits lock it in.
- **Paper 2**: closes the loop — **input demand, inverse demand, vessel × social-planner joint optimization** — to deliver welfare counterfactuals: equilibrium prices, species substitution & optimal quota allocation, and policy reforms.

> *Adaptive capacity is not a property of fishers; it is a property of the* ***bundle (portfolio × permit × institution × market)*** *they sit in. Identifying adaptation requires modeling all four dimensions jointly — pooling, fixed prices, or aggregate effort all bias the answer.*

# ¡Gracias! / Thank you

## Questions?

**Felipe J. Quezada-Escalona**

felipequezada@udec.cl

felipequezada.com

Departamento de Economía
Universidad de Concepción

Funded by ANID-Chile, FONDECYT Iniciación N° 11250223.

# Appendix

# Appendix A · Stock dynamics — spec & priors



**Process** (Schaefer with shifter on $r$):

$$B_{i,t+1} = B_{i,t} + r_{i,t} B_{i,t}\left(1 - \frac{B_{i,t}}{K_i}\right) - C_{i,t} + \varepsilon_{i,t}$$

with $r_{i,t} = r_i^0 \exp(\rho_i^{SST} \Delta SST_{t-1} + \rho_i^{CHL} \Delta \log CHL_{t-1})$.

**Observation** (log-normal): $\log B_{i,t}^{obs} = \log B_{i,t} + u_{i,t}$.

Three nested specs: **ind**, **omega**, **full**.

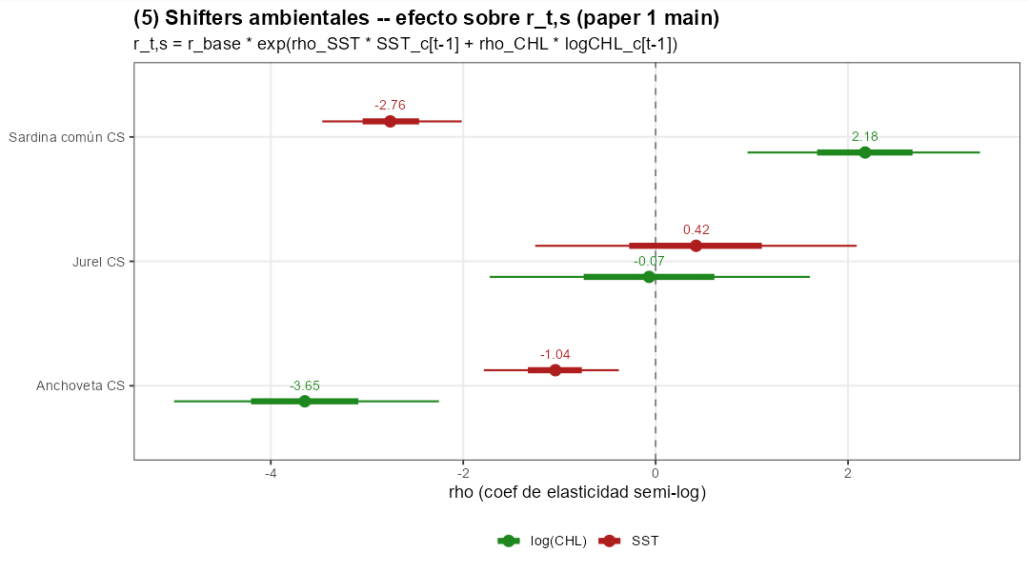
**Priors:**

- $r_i^0, K_i, \sigma_{proc}, \sigma_{obs}$ from IFOP / SPRFMO single-species assessments.
- $\rho_i^{SST}, \rho_i^{CHL} \sim \mathcal{N}(\hat{\rho}_i^{stress}, 1)$.
- $\Omega$: LKJ.

**Estimation:** Stan / HMC, 4 chains $\times$ 2000 post-warmup; $\hat{R} < 1.01$, $ESS > 400$.

# Appendix B · Climate shifters

Departamento de Economía
Universidad de Concepción




Posterior densities of $\rho^{SST}$ and $\rho^{CHL}$, full multi-species specification. Anchoveta and common sardine: priors dominated by data. Jack mackerel: posteriors track priors $\rightarrow$ **n.i.** and reported as such throughout.

# Appendix C · Stock productivity comparative statics



| Stock | Scenario | $\Delta r/r_0$ (cross-median) | 90% CI | $\Pr(\Delta < 0)$ |
|---|---|---|---|---|
| **Anchoveta** | SSP2-4.5, mid-century | **–51%** | [–68%, –27%] | 1.00 |
| | SSP5-8.5, end-century | **–90%** | [–98%, –54%] | 0.99 |
| **Sardine** | SSP2-4.5, mid-century | **–79%** | [–86%, –67%] | 1.00 |
| | SSP5-8.5, end-century | **–100%** | [–100%, –99%] | 1.00 |
| Jack mackerel | all scenarios | n.i. | n.i. | n.i. |

Cross-median across CMIP6 ensemble of 6 GCMs (CESM2, CNRM-ESM2-1, GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, UKESM1-0-LL); 90% CI is the within-posterior credible interval averaged across models.

# Appendix D · Why is jack mackerel n.i.? (triple evidence)



**(i) Spatial-domain robustness.** Re-fits over 3 alternative coastal domains (D1/D2/D3): $\sigma_{post}/\sigma_{prior}$ for $\rho^{SST}_{jurel}$ stays in [0.998, 1.014] — data adds no information regardless of domain. Anchoveta and sardine remain stable (CV $< 0.03$) across the same exercise.

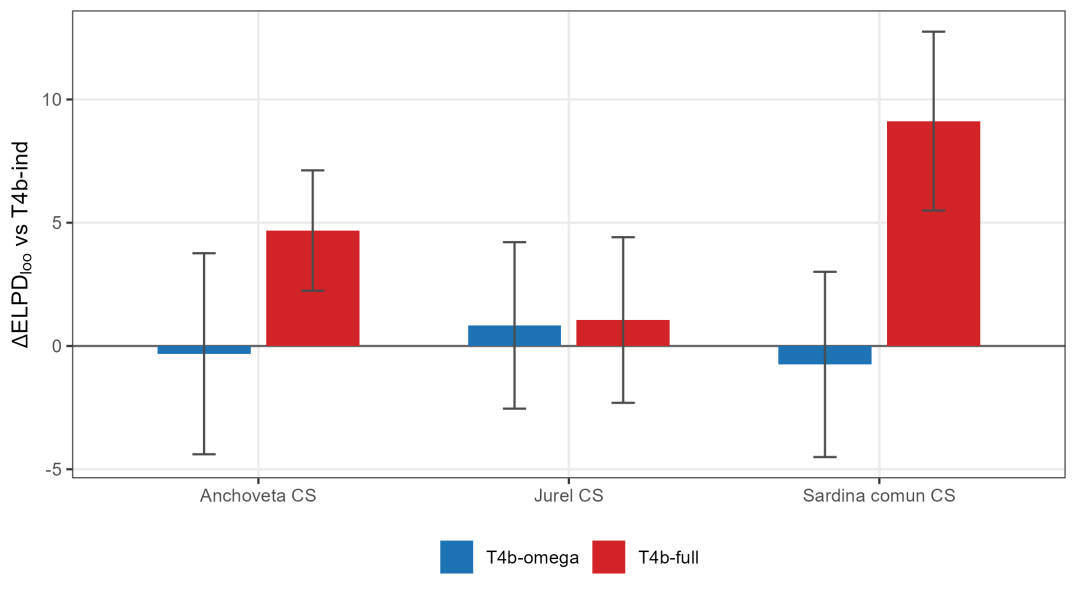
**(ii) Dual-source biomass.** Augmenting the CS state-space with the IFOP Northern Chile acoustic series (2010–24, $r^{log}_{CS,N} = 0.88$) does not break non-identification: $\sigma_{post}/\sigma_{prior} \in \{0.94, 0.99\}$ for $(\rho^{SST}, \rho^{CHL})$. The bottleneck is **structural**, not sample size.

**(iii) OROP-PS coherence.** SPRFMO transzonal index correlates with CS at $r = 0.11$ over $N = 17$ overlapping years. CS is decoupled from the basin-scale signal — range-wide forcing cannot pin it down either.

- Reduced-form linear stress tests on growth increments do **not** close the hindcast under SST/CHL forcing alone, consistent with structural non-identification.
- **Implication:** climate-driven projection for jack mackerel CS must come from a **transboundary, game-theoretic** model (cf. Session S9, bullet 2), not a CS-only structural fit.

# Appendix E · Model comparison (LOO)

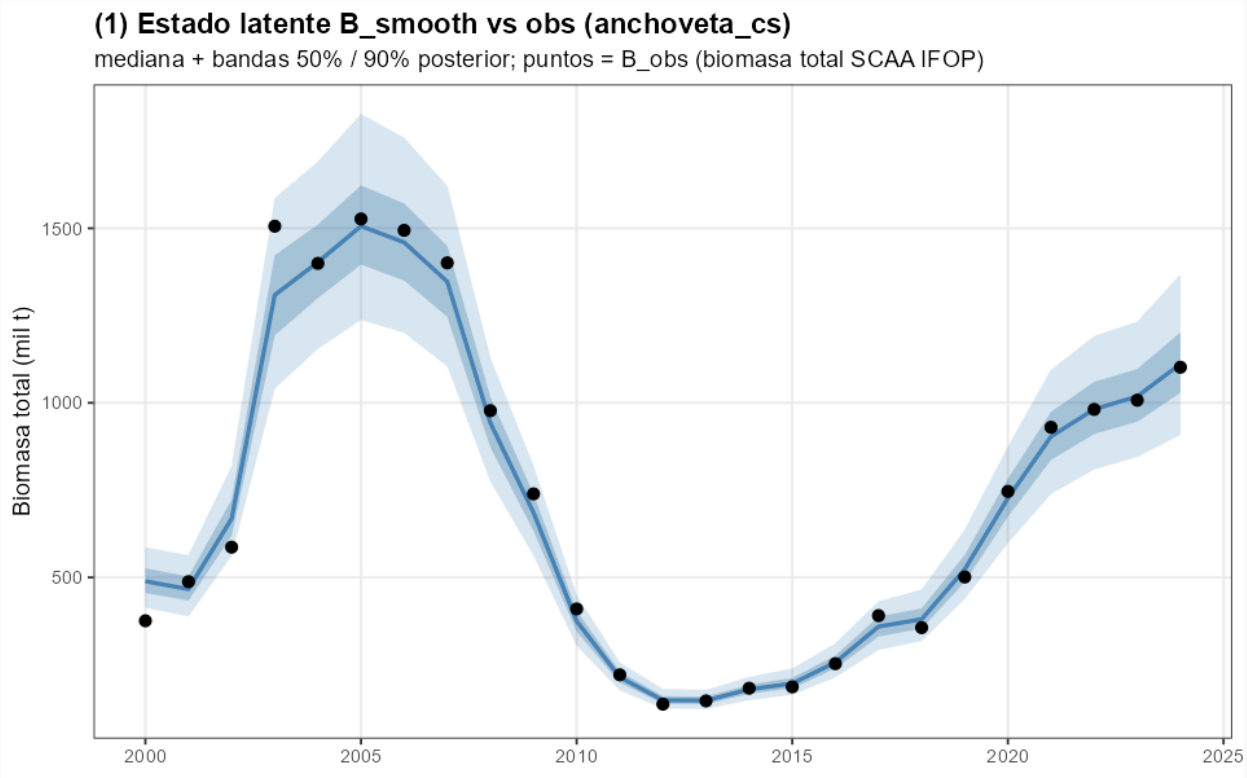





LOO ELPD across specs: **ind** ($\Omega$ diagonal, $\rho = 0$), **omega** ($\Omega$ full, $\rho = 0$), **full** ($\Omega$ full, $\rho \neq 0$). Full wins at $\Delta\text{ELPD} \approx 14$–$18$; LFO one-step-ahead essentially tied.

# Appendix F · Posterior predictive check

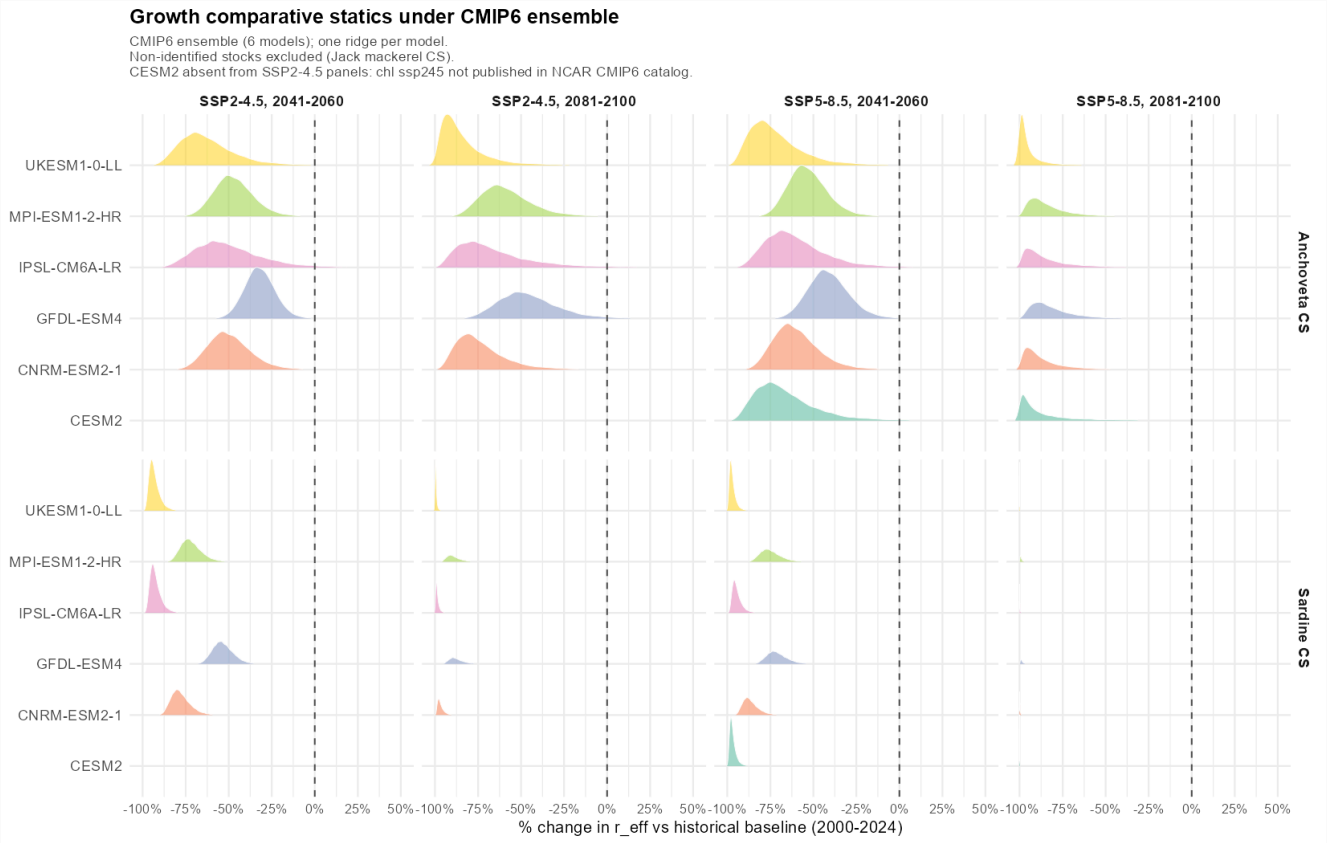
Departamento de Economía
Universidad de Concepción




Smoothed posterior of biomass vs observed acoustic estimates, full model.

# Appendix G · CMIP6 ensemble — ridgeline

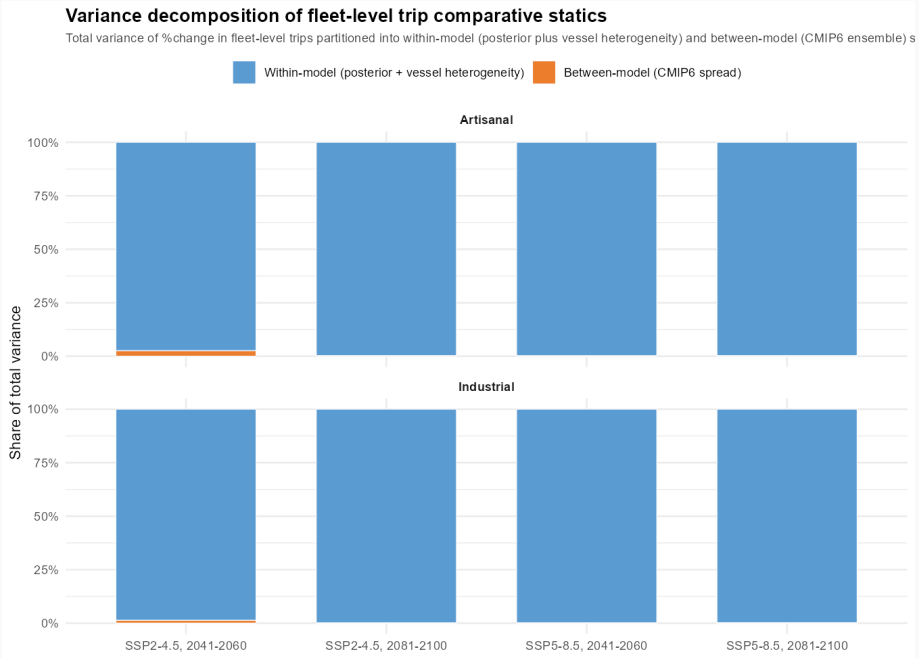





Posterior $\Delta r/r_0$ across SSP $\times$ period combinations, one ridge per CMIP6 model (6-GCM ensemble).

# Appendix H · Variance decomposition of trip response

Departamento de Economía
Universidad de Concepción




- **Within-model** (posterior + vessel heterogeneity) dominates: **97%–100%** across all SSP $\times$ period $\times$ fleet cells (re-computed after the direct weather channel was added).
- **Between-model** (CMIP6 ensemble): at most **3%**, often $\approx 0$.
- The narrow cross-IQR in Table 5 is *not* climate-model consensus — it is a **floor effect**: once $f_v^H \to 0$, $\%\Delta T \to \exp(-\beta_g H_v^{alloc})$ — a fleet-mechanical quantity, not a climate magnitude.

# Appendix I · Paper 2 — data inputs



- Trip-level catch and price data (IFOP logbooks, 2013–2024).
- All-gear official catch series (SERNAPESCA).
- Vessel characteristics & ports of operation.
- Environmental covariates (Copernicus, GlobColour, CMIP6 6-GCM ensemble).

> ***Caveat — IFOP panel coverage*** *covers the* ***purse-seine artisanal*** *subset; non-PS gears (lampara, lines) excluded.*